

Michael Cronin

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RELEVANT EXPERIENCE

Celestica <i>Data Engineering Intern</i> - Designed and migrated the supply chain database schema, cutting refresh latency by 70% and enabling near real-time KPIs. - Engineered automated ETL pipelines in Python/JavaScript, orchestrating data ingestion and transformation through Google APIs and enabling the seamless integration of new data sources across supply chain systems. - Developed and maintained 100+ PowerBI dashboards delivering predictive insights for executives and ML pilot programs.	May 2024 – Aug 2025 Toronto, ON
QMIND (Queen’s Machine Learning Club) <i>Director of Design</i> - Direct the Consulting Pod of 20 students delivering 4 client-partnered projects with firms in data, healthcare, and law, managing recruitment, project execution, and stakeholder relations to ensure industry-grade deliverables. - Lead a 5-person team developing a customizable content-repurposing agent, working with data consulting firm Project X.	Mar 2025 – present Kingston, ON
Queen’s University — Mathematics & Statistics Department <i>Undergraduate Researcher</i> - Developed a control framework proving stability theorems across time scales focused on energy-efficient controller design. - Awarded an NSERC Undergraduate Research Award (\$6,000+).	May 2023 – Aug 2023 Kingston, ON
Queen’s Engineering Society <i>Director of Design</i> Oversaw 20 engineering design teams (up to 200 members each), steering strategy, funding, and cross-team collaboration.	May 2022 – May 2023 Kingston, ON

SELECT PROJECTS (SEE ALL)

Compile: AI News Intelligence Platform (Live Site) <i>Next.js, Python, Supabase, Pinecone, Temporal, Crawl4AI, Vercel</i> - Built an AI news and media aggregator that curates articles based on user preferences and interests. - Ingests 500+ articles daily via a self-healing Crawl4AI pipeline; stores content in Supabase and Pinecone for real-time access. - Clusters related stories semantically and generates personalized summaries using Gemini 2.5. - Developed a React + Next.js frontend with semantic search, “For You” feeds, recommendations, and tone/length rewriting.	2025
Mantis: Web UI Autotester (Hack The North Finalist - Best Dev Tool) (Github, Devpost, PyPI) <i>Python, Flask, Cohere, PyPI Package, Playwright</i> - Developed an open-source web crawler to detect and report UI bugs and accessibility issues, deployable via simple CLI. - Packaged tool as a PyPI module and showcased at Hack the North; supports custom depth, page limits, and JSON output.	2025
Chameleon (HackAI Winner) (Github, Devpost) <i>Next.js, Python (FastAPI, LangGraph), Supabase</i> - Won “Best AI Retrieval” at HackAI Toronto by transforming long-form videos/blogs into creator-styled LinkedIn/X posts. - Designed a retrieval agent with LangGraph + Gemini; deployed a FastAPI backend and Next.js frontend on Vercel/Supabase.	2025
Mechanistic Interpretability for LLM Jailbreak Defense (Research Paper) <i>Python, SAELens, Gradio</i> - Developed a framework that improved LLM resilience to jailbreak prompts (40% higher refusal rate) while sustaining performance on benchmark tasks. - Co-authored a paper published at CUCAI (Canadian Undergraduate Conference on Artificial Intelligence).	2024 – 2025

EDUCATION

Queen’s University — Smith School of Engineering and Applied Science <i>B.A.Sc., Applied Mathematics & Computer Engineering; GPA: 4.18/4.30</i> - Relevant Coursework: Data Structures, Algorithms, Object-Oriented Programming, Databases, Deep Learning - Achievements: Jeffery Award in Mathematics [2024] for top standing in class - \$23,400; further scholarships valued at \$10,000	Kingston, ON Class of 2026
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ADDITIONAL INFORMATION & INTERESTS

- Programming Languages & Tools:** Python (TensorFlow, Keras, Pandas, NumPy), Java, C, C++, JavaScript/TypeScript (React, Next.js, Node), MATLAB, SQL; Supabase, PowerBI, Google APIs, Git, Vercel, FastAPI, LangGraph, RAG, REST APIs.
- Active Volunteer:** Five EngSoc roles including Orientation Week; recognized as top extracurricular contributor in class.
- Outdoors Enthusiast:** Led canoe trips (26-day white-water expedition; >800km paddled); avid skier and tennis instructor.